

REVERSE INNOVATION, EMERGING MARKETS, AND GLOBAL STRATEGY

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'Reverse innovation' refers to the case where an innovation is adopted first in poor (emerging) economies before 'trickling up' to rich countries. Although examples of reverse innovation are still rare, it raises interesting theoretical questions, such as what kinds of innovation emerging economies are likely to spawn, why such innovations might diffuse to rich countries, what competitive advantages local and foreign firms enjoy in this process, and how it affects the global strategy and organization of established MNEs. Research on reverse innovation can enrich and extend mainstream theories of innovation, internationalization, MNE management, and FDI spillovers. Copyright © 2011 Strategic Management Society.

INTRODUCTION

The locus of innovation in the global economy appears to be changing because of the rise of emerging economies, such as China and India, and the 'flattening' of the world. Poor, emerging markets no longer just borrow innovations from developed countries; from time to time they also contribute innovations to the rest of the world, including developed countries.¹ We refer to this as reverse innovation, that is, the case where an innovation is adopted

first in a poor country before being adopted in rich countries. Instances of reverse innovation still appear to be rare, and it is hard to tell if this will change materially in the future. Therefore, this article is about a nascent phenomenon whose future potential is uncertain. Nevertheless, reverse innovation is a promising area for research by international business and strategy scholars because it provides the opportunity to enrich and extend mainstream theories in a number of areas.

This article takes the first steps in that direction. It proposes a research agenda that explores the nature of innovation in emerging markets, the reasons why such innovations might diffuse to other parts of the world, the role of local firms and developed-country multinational enterprises (DMNEs) in this process, and the strategic and organizational challenges faced by DMNEs that pursue reverse innovation. In each of these areas, the article offers illustrative hypotheses and questions for further research. However, these are only tentative ideas based on assumptions and arguments that need to be evaluated and tested rigorously.

Keywords: reverse innovation; glocalization; emerging markets; emerging-market multinationals (EMNEs); local growth teams; product cycle theory

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¹ We use 'emerging markets,' 'developing countries,' and 'poor countries' interchangeably and likewise use 'industrialized countries,' 'developed countries,' and 'rich countries' interchangeably. The latter include the Triad markets of North America (U.S., Canada), Western Europe, and Japan, while the former include the rest.

Research on reverse innovation can extend mainstream theories in at least four areas—innovation, internationalization, multinational enterprise (MNE) management, and foreign direct investment (FDI) spillovers. Most of this article is devoted to illustrating such opportunities for theory building. In addition, research in this area should have practical relevance for managers and policy makers if reverse innovation becomes more widespread in the future.

WHAT IS REVERSE INNOVATION AND WHY STUDY IT?

Innovation has always been recognized as one of the pillars of multinational firms (Bartlett and Ghoshal, 1989). In the mainstream international business (IB) literature, innovation often is equated with technological innovation (Caves, 2007) and is assumed to originate in the developed countries where the world's leading multinational corporations (MNCs) are located (see Figure 1). In the original product cycle hypothesis, for instance, Vernon (1966) saw the United States after the end of World War II as the technology leader and the source of product innovations that targeted high-income consumers and process innovations that substituted capital for labor. He argued that U.S. multinationals spread both types of innovation to Europe and Japan and, eventually, developing countries. By the 1970s, Europe and Japan had closed their technology and income gaps with the U.S., and Vernon (1979) argued that innovations would now diffuse horizontally among developed countries and downward to developing countries.

Subsequent research has recognized that DMNEs strive to capture knowledge and innovation in many locations around the world, not just their home countries (Cantwell and Mudambi, 2005; Kuemmerle 1997), but as late as 2004, DMNEs had 90 percent of their foreign R&D affiliates in the Triad regions and only 10 percent in developing countries (United Nations, 2005). The possibility of important innovations occurring in developing countries and then trickling up to developed countries is only now getting recognition (e.g., Immelt, Govindarajan, and Trimble, 2009; Ramamurti 2009a).

Well-known examples of recent innovations in emerging markets include the Tata Nano, Grameen Bank (microfinance), GE's ultrasound, Embraer's regional jets, BYD's electric cars, Bharti Airtel's ultracheap wireless telephony, and Nokia's inexpensive cell phones. A few of these innovations have already made the journey from poor to rich countries. But innovation in poor countries seems to be deeper and wider than these examples indicate, and more of those innovations may diffuse internationally in the future. *The Economist's* special issue on innovation in emerging markets, subtitled 'The world turned upside down,' noted that 'the emerging world, long a source of cheap labor, now rivals the rich countries for business innovation' (*The Economist*, 2010: 1). After citing examples from many industries, it concluded that these innovations will change 'not just emerging markets but the rest of the world as well' (*The Economist*, 2010: 16).

Innovations occurring in emerging economies tend not to involve technological breakthroughs of the kind that drive innovation in developed countries.

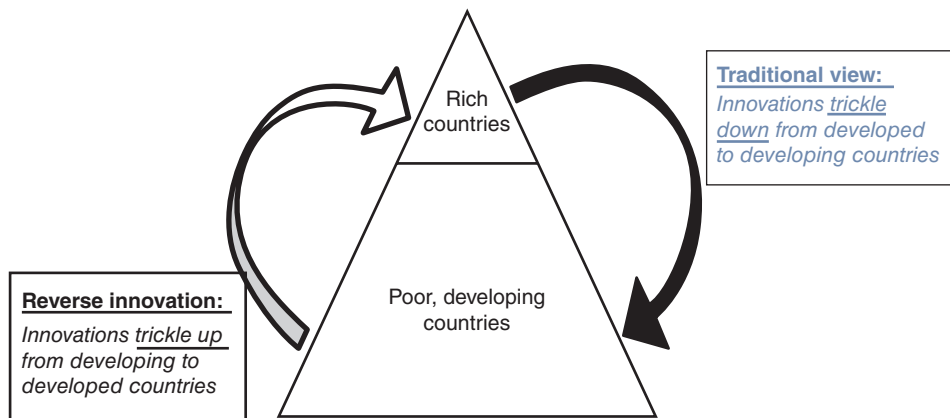


Figure 1. Reverse innovation

They do, however, involve novel and innovative combinations of existing knowledge and technologies to solve pressing local problems and the use of new processes and business models. The firms leading these innovation efforts could be local firms or foreign MNEs with operations in emerging markets. Also, innovations first adopted in a developing country may draw on talent, technology, and ideas from many parts of the world, as is to be expected in today's integrated world. General Electric's ultralow cost ultrasound and ECG machines, for instance, were led by development teams in China and India, initially for use in these countries, but with inputs from local and foreign subsidiaries. The same was true of the development of the Tata Nano car. Our focus in this article is on where an innovation is first *adopted*, not where it is *developed*. Likewise, the adjective 'reverse' refers specifically to the flow of innovations from poor to rich countries, which is opposite to the predominant direction of technology flow in modern times.² It is not meant to connote any other kind of value judgment.

Compared to the flow of innovations and technology from rich to poor countries, flows in the opposite direction probably are still minuscule. Indeed, we believe more research and case studies are needed to establish the true extent of reverse innovation and its future potential. Our claims here are relatively modest—that innovative activity in emerging economies is much greater than it has ever been, that it is growing rapidly, and that the resulting innovations may increasingly find their way to the rest of the world, including developed economies.

In its totality, reverse innovation, as described here, seems to be without historical parallel, at least in recent times, but many of its constituent ideas do have historical parallels and have been researched extensively by strategy and IB scholars. The latter include studies of how location affects innovation (Porter, 1990; Nelson, 1993), why firms innovate outside the home country (Kuemmerle, 1997), how and why innovations diffuse across countries (Vernon, 1966, 1979), how firms adapt products designed for one country for sale in other countries (Yip, 1989), and how MNEs manage the tension between global standardization and national responsiveness (Prahalad and Doz, 1987; Bartlett and

Ghoshal, 1989). Also familiar is the notion of firms in industrially backward countries playing catch-up with First World firms, typically starting with low-end products and then working their way up to high-end products, e.g., Japanese or Korean firms in the postwar period (Tsurumi, 1976; Amsden, 1992). For these reasons, extant theory can help us understand many aspects of reverse innovation. For example, the notion that innovations occur in response to opportunities and constraints in a firm's immediate business environment seems to be just as true in poor countries as in rich countries; it's just that the opportunities and constraints in poor countries are systematically different from those in rich countries. It would be a mistake, however, to conclude that because some facets of reverse innovation have a familiar ring or are well understood, there is nothing new about it in whole or in part and that there is no need to rethink some mainstream theories.

DMNEs may have innovated outside the home country in the past, but they seldom did so in really poor, developing countries, and they were seldom joined in this effort by entrepreneurial local firms, as is occurring today.³ Similarly, DMNEs may always have adapted products designed for one Triad market to suit other Triad markets, but that is not the same as adapting Triad-based products for the mass market of poor countries. True, Japan and Korea also had poor consumers when they began to catch up after World War II, but the income gap between them and the U.S. was much smaller than what exists today between China or India and the U.S. For instance, in 1985, after more than three decades of economic recovery, Japan's per capita income in purchasing power parity (PPP) terms was 81 percent of the U.S. level (Balassa and Noland, 1988). In 2009, fully 30 years after China's economy opened up, its per capita income in PPP terms was only 15 percent of the U.S. level; India's was only 7 percent of the U.S. level.⁴ The scale and size of the Chinese and Indian markets, each potentially as big as the U.S., but much poorer, is also without parallel in recent

² To be sure, in the pre-Industrial Revolution era, important innovations did flow in the other direction, e.g. Chinese inventions such as gunpowder, silk, or paper found their way to the West.

³ One admittedly imperfect indicator of growing innovation capability in poor countries is the rapid growth of patents filed by firms from these countries. *Fortune* reports, for instance, that in 2011 China is likely to file more patents than the U.S. or Japan (Tseng, 2010). Although the quality of these patents is suspect, the growth in numbers is indicative of China's growing importance as a source of innovations.

⁴ Japan figures from Balassa and Noland (1988). In 2009, per capita income in PPP terms in China was only \$6,778 and in India only \$3,015, compared to \$45,934 in the U.S. (See IMF, 2010.)

Table 1. Mainstream theories and reverse innovation

Topic	Mainstream view	Observed aspects of reverse innovation that do not fit with mainstream view	Specific questions explored in article
Innovation	1. Innovations trickle down from rich to poor countries 2. Innovations begin with lead users and spread to others	1. Innovations may also occur in poor countries and 'trickle up' to rich ones 2. Reverse innovations may begin with laggards rather than lead users	• What kinds of innovations do EMs spawn? • Why is this occurring now and not earlier? • Why would innovations 'trickle up'?
Internationalization Process	1. MNEs originate in rich countries and spread globally, including to poor countries, based on proprietary technologies/brands	1. MNEs also originate in EMs and spread globally, including to developed countries, leveraging home-based innovations	• What are the competitive advantages of EM firms and how do these facilitate internationalization? • How do EMNEs and DMNEs compete in the different stages of reverse innovation?
DMNE strategy and management	1. DMNEs develop global product platforms in Triad markets and adapt them for EMs 2. DMNEs give global product mandates to subsidiaries in other Triad markets	1. DMNEs develop new product platforms in EMs for EMs 2. DMNEs give global mandates for some products to subsidiaries in EMs	• How does reverse innovation affect the 'glocalization' strategy of DMNEs? • Can DMNEs pursue glocalization and reverse innovation at the same time?
FDI spillovers	1. Local firms in EMs capture spillovers from DMNEs that invest in EMs	1. DMNEs capture spillovers from local firms when they invest in EMs (i.e., learning is bidirectional)	• Who learns from whom and why, when EMNEs and DMNEs come into contact in different markets?

Note: EM = Emerging markets; EMNEs = Emerging-market MNEs; DMNEs = Developed-country MNEs.

history (Wilson and Purushothaman, 2003). Likewise, Japanese and Korean companies did export cheaper and better cars to the U.S., disrupting the American auto industry, but that is not the same as producing ultralow cost products, such as a \$2,500 car or a \$50 computer. Further, Western multinationals were unable to enter and compete in postwar Japan and Korea due to tariff, nontariff, and FDI barriers, but this has been less true of post-liberalization China or India. As a result, DMNEs have been able to participate in the innovation process of emerging economies in ways they could not in Japan or Korea. And, finally, the world itself has changed profoundly in the last 30 to 40 years, becoming much 'flatter' than before (Friedman, 2005), with 'gateways to internationalization' that did not exist earlier (Williamson and Zeng, 2009). In summary, reverse innovation is a phenomenon with familiar features *as well as* new features, and it is

occurring in a changed world compared to even the 1960s and 1970s.

Table 1 illustrates the theoretical challenges (or opportunities) reverse innovation presents for mainstream theory in four important areas: innovation, internationalization process, DMNE strategy and management, and FDI spillovers. The second column caricatures the mainstream view on each topic; the third column shows why reverse innovation fits awkwardly, or not at all, with that view; and the last column lists specific research questions explored in this article. These questions are by no means comprehensive, and the answers proposed in this article are merely initial guesses intended to provoke further inquiry. To reiterate, the table purposely highlights areas in which reverse innovation challenges, rather than agrees with, mainstream ideas because areas of challenge can do more to advance extant theory.

A RESEARCH AGENDA

The rest of this article presents a research agenda for each topic area in Table 1.

Revisiting innovation

Reverse innovation poses interesting puzzles for extant theory about where innovations occur, why, and how they diffuse internationally (Table 1). One question is why poor countries might be centers of innovations and what kinds of innovation they might spawn—regardless of whether those innovations trickle up to rich countries. A second question is why innovation is flowering in emerging markets today but did not in earlier decades. And a third question is when and why innovations might move in the counterintuitive direction from poor to rich countries. After all, if the large ‘distance’ (Ghemawat, 2001) between poor and rich countries is what makes local innovation necessary in emerging markets, shouldn’t the same reason make it harder to transfer the resulting innovations to rich countries?⁵

What kinds of innovations will emerging markets spawn?

One important line of research is to understand the type of innovations emerging markets are likely to spawn, drawing on the literature on national innovation systems (Nelson, 1993; Porter, 1990). The most important dimension on which emerging economies appear to differ from developed economies is the per capita income of average consumers, except perhaps at the very top of the economic pyramid. Therefore, the mass market in poor countries requires products with dramatically improved price-performance features, creating opportunities for ‘affordability innovation.’

Success in these markets also requires optimizing products to suit local conditions, such as making them sturdy to compensate for the harsh conditions under which they are typically used, making them portable so equipment can be taken easily to far-flung customers whose mobility is limited, or making them easy to operate and maintain because users are technically unsophisticated. Because of institutional voids of various kinds, innovations may also be necessary in the way products are sold,

distributed, and financed (Khanna and Palepu, 2005). All this adds up to a fundamental rethinking of the business model necessary for success in emerging markets (Ramamurti and Singh, 2009b).

In other words, large differences in per capita income between poor and rich countries and the conditions under which products are used seem to provide the basis for innovations in poor countries. Ghemawat (2001) captures some of these differences in his multidimensional measure of cultural, administrative, geographic, and economic distance between countries. However, much more research is needed to understand how such ‘distance’ creates pressure for local innovation in emerging markets or how it inhibits their diffusion to developed countries.

Reverse innovation can also build on the literature on lead users and their role in diffusing innovations. Von Hippel (1986) views lead users as early adopters of cutting-edge innovations who have demanding standards and are not very price sensitive. Firms work with them to launch new products that later can be targeted at mainstream consumers. In this framework, the left tail of the distribution consists of laggards—individuals who are extremely value conscious, happy with ‘good enough’ quality, and costly to reach and serve, a profile that resembles the mass market consumer in emerging economies. In other words, just as lead users can be leveraged to diffuse an innovation to mainstream users, reverse innovation can be used to turn laggards into mainstream users. Future research can examine such issues as the conditions under which laggards, rather than lead users, will be a more advantageous source of new product ideas and whether some industries are more likely than others to benefit from reverse innovation.

Why now and not earlier?

Another interesting question is why innovation may be flourishing in emerging economies now and not in prior periods. This would require historical analysis of the national and international environments at different times to discern key driving forces. On this question, our tentative guess is that the demand for and supply of local innovation have only just come together in emerging markets. On the demand side, we referred earlier to accelerating growth in emerging economies and the simultaneous slowing down of developed economies. In the coming decades, as much as two-thirds of world GDP growth is likely to occur in these markets. The needs of middle class consumers in emerging markets are, therefore, of

⁵ We are grateful to Juan Alcacer for raising this very interesting question.

strategic significance to firms everywhere. For local innovation to actually occur, however, demand for such innovation must be matched by supply, and it is on this front that conditions have changed dramatically in the last two decades. Economic liberalization in many emerging markets, coupled with a 'flatter' world brought on by technological change (Friedman, 2005), have spawned local firms in emerging economies that can draw on local and global resources to innovate for local markets (Williamson and Zeng, 2009; Ramamurti, 2009a). An innovation like Bharti Airtel's telecom revolution could not have happened in pre-liberalization India or without the access to international capital, suppliers, and technology made possible by a flatter world (Martinez-Jerez, Narayanan, and Jurgens, 2006). At the same time, DMNEs have realized that it is not enough to serve just the premium segment in emerging markets with products that are close variants of those developed for rich countries. Instead, they have had to develop products with drastically lower costs and special features, which were not always possible simply by tinkering with existing designs (Prahalad, 2005).

In theory, DMNEs have long had access to the technology and skills needed to create something like the Tata Nano but, in practice, there was little incentive for them to do so. Without a local firm taking the lead, the latent demand of Indians for such products probably would have remained unrealized. However, once local firms developed such products, DMNEs were drawn into the game out of fear those firms eventually would exploit their innovations globally and challenge DMNEs everywhere (Ramamurti, 2009a). This has added momentum to the supply side of local innovation in emerging markets.

Why might innovations 'trickle up' from poor to rich countries?

Reverse innovation entails at least three stages. The first is adoption of an innovation in one emerging market, such as China or India. The second is the transferring of this innovation to other emerging markets. The third and final step is transferring it selectively to developed-country markets. Both local firms and DMNEs can engage in all three stages, but their strengths and weaknesses are likely to be quite different, as will be discussed. Because the local firms that take such innovations abroad are, by definition, multinationals, we refer to them as 'emerging-market multinational enterprises' (EMNEs).

The diffusion of innovations from one emerging market to another with similar circumstances is not particularly surprising (Stage 2) because both may share customers who are price sensitive or need products that are portable, rugged, and easy to use. But the trickling up of innovations from poor to rich countries (Stage 3) is perplexing. If 'distance' between countries (Ghemawat 2001) makes it necessary to innovate specifically for emerging markets, shouldn't the same distance factors make it harder for the resulting innovations to migrate to rich countries? Furthermore, the product cycle theory provides a persuasive argument for why innovations will diffuse from developed to less developed countries as the latter become rich, but since rich countries seldom become poor, why would innovations ever travel in the opposite direction?

We propose five possible reasons why innovations might trickle up from poor to rich countries (see Table 2). First, there are poor people in rich countries for whom 'good enough' products at ultralow prices may have great appeal. This is the mirror image of the argument that products developed for rich countries may have an immediate market in poor countries because there are rich people in poor countries. Thus, innovations in low-cost housing, medical care, or banking (e.g., microfinance) may appeal to the inner city or rural poor in developed countries. Admittedly, this may be a thin slice of the market and may not be considered attractive enough by some DMNEs, but it could be a nice avenue for growth in otherwise saturated businesses, and it could become more important if incomes stagnate in the rich countries.

Second, products that have been redesigned to be ultralow cost for emerging markets may expand overall market demand in developed countries. Because the price reductions necessary to succeed in emerging markets are dramatic (on the order of 70 to 90 percent of relative prices in developed countries), demand may be expanded significantly because of the price elasticity effect when new versions are introduced in developed countries at these ultralow price points. It is hard to imagine that the demand for many products would not expand dramatically if prices were cut to 10 or 20 percent of what they used to be. There are numerous historical examples of large expansion in demand because of falling costs and prices, fueled by the so-called 'experience curve effect'—e.g., televisions, videocassette recorders, copiers, personal computers, and so on (Hall and Howell, 1985). Because the cost reductions

Table 2. Why innovations may trickle up from poor to rich countries

No.	Trickle-up mechanisms	Examples
1	Innovations developed in EMs may have a ready market among poor people in rich countries.	■ Microfinance, invented in Bangladesh, works also for the inner city poor in rich countries like the U.S.
2	Dramatic cost and price reductions of 70 to 90 percent achieved to succeed in EMs can help expand demand in rich countries.	■ Dropping prices to 15 percent of original levels expanded demand for GE's ultrasound machines in the U.S.
3	New features incorporated for EMs, such as sturdiness, portability, or ease of use, may create new market segments in rich countries.	■ Making ECG machines portable and more compact created new market segments for GE in the U.S.
4	Technology of 'good enough' products developed for EMs may improve over time to satisfy high-end applications in rich countries.	■ Portable ultrasounds developed by GE for China were later useable for high-end radiology and obstetrics applications.
5	EMs may leapfrog to latest technologies, especially if they have large internal demand, are unencumbered by legacy technologies, and face fewer regulatory obstacles.	■ EMs have advanced capabilities in industries such as wireless banking, nonconventional energy, and electric cars, which have relevance and value in developed-country markets.

Note: EM = Emerging market.

demand by emerging markets are an order of magnitude greater than those resulting from the experience curve effect, the potential expansion of demand in developed countries may be that much greater as well.

Third, as shown in Table 2, optimizing products for emerging markets requires adding new functionalities, such as portability or ease of use, that may also create new applications in developed countries. For instance, making a product sturdier or easier to use may expand its appeal to at-home users in rich countries. Thus, redesigning existing products from the ground up for emerging markets may help generate a second round of growth in those products in developed markets.

Fourth, research on disruptive innovations (Christensen and Bower, 1996) argues that over time, technology of low-cost products can be improved until these products satisfy the more demanding mainstream customer. Thanks to advances in technology, for instance, GE's portable ultrasounds, originally developed in China, were able by 2009 to perform radiology and obstetric functions that once required high-end conventional machines (Immelt *et al.*, 2009).

Finally, emerging markets can also be at the forefront of global innovations when they leapfrog over legacy technologies to adopt frontier technologies. Examples include the leap in many poor countries from no telephone or obsolete electromechanical

exchanges to the latest in wireless telephony. That revolution, in turn, is causing poor countries to leapfrog past brick-and-mortar banking to wireless banking for the masses. A similar dynamic may be underway in the adoption of nonconventional energy sources, such as wind and solar, led by emerging-market multinationals like Suzlon Energy Limited and Goldwind Science and Technology Co., Ltd. (which are among the 'global first movers' in their industries) (Ramamurti and Singh, 2009b; Williamson and Zeng, 2009). In all these examples, the large potential market in poor countries, unencumbered by sunk investments in old technologies or complex regulatory systems, may allow these countries to move faster than rich countries. But the solutions they develop are likely to have value and relevance for rich countries.

For all these reasons, it is plausible that innovations will trickle up from poor to rich countries. Reverse innovation, thus, provides an opportunity to update Vernon's (1966, 1979) product cycle theory to explain such things as ultralow cost innovations in emerging markets and the trickling up of innovations. It also raises interesting questions about the circumstances under which it is more likely to happen. For instance, is it more likely in some industries than others, or between some country pairs than others? These empirical questions will benefit from carefully conducted case studies and large sample studies if and when sufficient examples become available.

Revisiting internationalization process

As shown in Table 1, the literature on internationalization process has been based largely on the experience of firms from developed countries, although this is starting to change (Aulakh, 2007; Luo and Tung, 2007; Ramamurti and Singh, 2009a; Guillen and Garcia-Canal, 2009). As reverse innovation is often triggered by emerging-market firms, a question arises: what competitive advantages, or 'ownership advantages,' do these firms enjoy that allow them to pioneer innovations, sometimes even ahead of DMNEs? A second interesting question is whether these ownership advantages and capabilities can be the foundation for successful internationalization. There is a debate in the literature about whether EMNEs go abroad to exploit existing ownership advantages or acquire new advantages (Mathews, 2002; Narula, 2006). Clearly, there is a need for more research into the ownership advantages of EMNEs, including a willingness to consider new ownership advantages that are not usually associated with DMNEs.

What are the competitive advantages of emerging-market firms?

Some observers have argued that EMNEs rely too much on home country advantages for their international competitiveness (Rugman, 2009) and that they possess only 'ordinary resources' (Madhok, 2010) or few intangible ownership advantages, such as brands and cutting-edge technology that can sustain long run performance. Others disagree, pointing to a variety of ownership advantages possessed by EMNEs (Guillen and Garcia-Canal, 2009; Lessard and Lucea, 2009; Williamson and Zeng, 2009). EMNEs are seen as having a deep understanding of the needs of local customers, strengths in ultralow cost design and manufacturing, and strong distribution in their home markets—all of which translate into significant competitive advantage in many 'mid-tech' industries, not only in the local market but also in other emerging markets (Ramamurti, 2009a, 2009b). Occasionally, as noted earlier, EMNEs may even be at the forefront of an emerging industry, acting as 'global first movers' (Ramamurti and Singh, 2009b). EMNEs pursuing either strategy are likely to trigger Stage 1 of the reverse innovation process and then participate in the international diffusion of those innovations. In the future, more DMNEs may trigger Stage 1 themselves to preempt local firms from getting a head start with reverse innovation.

As they innovate at home or expand abroad, EMNEs are likely to encounter competition from DMNEs, which raises the interesting question of how their relative competitive advantage is likely to play out during the three stages of reverse innovation.

How do EMNEs and DMNEs compete in the diffusion of reverse innovations?

Table 3 compares the competitive advantages of EMNEs and DMNEs in each stage of reverse innovation. Once again, these are only educated guesses informed by anecdotal evidence, and they require more rigorous development and testing in the future. In Stage 1, both EMNEs and DMNEs have significant strengths, but in different areas. EMNEs are likely to have a deeper understanding of local customers and their preferences, as well as stronger local brands and distribution through which local innovations can be exploited. The technological capabilities of EMNEs are apt to be narrower and shallower than those of DMNEs, but they are likely to have the capacity to optimally adapt foreign products and technologies for local markets. In so doing, they are also likely to drastically reduce capital and operating costs, resulting in ultralow cost products and services (Ramamurti, 2009a). EMNEs are less constrained by prior investments or market positions that need to be protected while pursuing innovations for the home market. As local players, EMNEs are likely to be well connected to important local actors, including the local and state governments. With ownership typically in the hands of family business, they are likely to take a long-term view and be patient in winning in the home market.

Unlike DMNEs, EMNEs live or die by their home markets and will, therefore, persevere and adapt as needed to succeed. They usually are quicker to make strategic decisions and nimbler than the local subsidiaries of DMNEs that require approvals from remote headquarters (Amsden, 2009). DMNEs will also be held back by the need to protect past investments and market positions that could be disrupted by radical innovations for emerging markets. However, DMNEs have a deeper reservoir of technological capabilities, deeper pockets to fund innovation, and the prospect of a bigger market in which new innovations can be exploited because of positions already held in markets around the world. Overall, we suspect the scales are tipped in favor of local firms engaging in innovations for emerging markets. Without prodding from a local firm,

Table 3. EMNEs vs. DMNEs in the three stages of reverse innovation

Stage	Main challenges for DMNEs	Relative advantages of DMNEs	Relative advantages of emerging-market MNEs (EMNEs)
1 Winning in key emerging markets	• Zero-based innovation for a foreign market • Giving the emerging-market subsidiary access to firm's global technology	• Technology reservoirs within the firm • Familiarity with several emerging markets • Deep pockets	• Customer intimacy • Flair for low-cost solutions • Clean slate approach because of fewer prior investments • Strong commitment to local market • Access to local resources and capabilities • Patient capital
2 Winning in other emerging markets	• Managing transfer to other EMs	• Preexisting distribution and brand recognition in EMs	• Product pricing and features better suited to emerging markets
3 Winning in developed-country markets	• Positioning the reverse innovation vs. existing offerings • Managing the risk of cannibalization	• Strong presence, customer intimacy, brand recognition, and distribution in rich country markets	• Unconstrained by prior investments or risk of cannibalization • Prospect of rising margins in moving upscale • No internal resistance to expansion

Note: EMNEs = Emerging-market MNEs; DMNEs = Developed-country MNEs.

DMNEs may not deviate from their existing business models; it is the actual or imminent threat of disruption by local firms that prompts DMNEs to pursue reverse innovation.

In Stage 2, however, the EMNE and the DMNE may be equally matched in their ability to transfer innovations from the country of origin to other emerging markets. The former is likely to have products and services better suited for emerging markets and feel more comfortable operating in the difficult environment that prevails in emerging markets (Cuervo-Cazurra and Genc, 2008). However, the DMNE is likely to have a stronger preexisting market position in emerging markets (brands, distribution, customer relationships, etc.) that it can leverage to exploit the innovation.

In Stage 3, the DMNE is likely to have a definitive edge over the EMNE in transferring the innovation to developed countries. Now the shoe is on the other foot. The DMNE is more likely than the EMNE to better understand customers in its traditional (i.e., developed country) markets and it is likely to be strongly entrenched in these markets. Transferring the innovation will not require creating a new subsidiary or making expensive acquisitions, as might

be the case for the EMNE. But DMNEs will have to contend with strong internal resistance to transferring radical innovations from emerging markets because such innovations have the potential to cannibalize existing sales and profits in traditional markets and might adversely affect the firm's brands and image. It is not clear what the optimal strategy is for a DMNE to manage this process of 'self-disruption,' and this is one of many areas in which future research is needed. However, even DMNEs that are not enthusiastic about Stage 3 of the reverse innovation process may find themselves having to engage in the first two stages just to succeed in high-growth emerging markets.

In summary, EMNEs seem to have the edge over DMNEs in the first stage of reverse innovation, but the advantage seems to shift progressively toward DMNEs in the later stages.

Revisiting DMNE strategy and organization

Prior research has argued that global strategy of DMNEs can be viewed as creatively resolving the tension between economies of global scale and the advantages of local adaptation (Prahalad and Doz,

1987; Bartlett and Ghoshal, 1989; Ghemawat, 2007). Since the overall approach combines the best of globalization and localization, it can be labeled ‘glocalization.’ Although reverse innovation may appear to be a special case of a familiar problem (i.e., how to balance national responsiveness and global standardization), the amount of national responsiveness necessary to compete in the mass markets of poor countries may be considerably greater than that required to adapt products from one Triad market to another.

GE’s struggle and subsequent success in penetrating the market for medical diagnostic imaging equipment in China (see Appendix 1) suggests one fruitful line of future inquiry: how can DMNEs pursue glocalization and reverse innovation simultaneously? Glocalization centralizes strategy formulation in global product divisions headquartered in the rich countries, and subsidiaries in emerging markets play implementer roles, with mainly sales and distribution functions. Glocalization has been and is a useful and necessary approach, but DMNEs might struggle with reverse innovation precisely because they are so good at glocalization. The centralized, product-focused structures that have made multinationals so successful at glocalization could create three traps that might make it difficult for the DMNE to seize opportunities in emerging markets.

Familiarity trap

Organizations favor the familiar (Ahuja and Lampert, 2001; Christensen and Bower, 1996). The global product division headquartered in a rich country understands customer needs in the rich world. It is very difficult for the same unit to identify opportunities for business model innovations in emerging markets. Emerging markets, after all, are two to three levels below the CEO and do not get the mind share, voice, or visibility of the bigger, more developed markets. For the global product division, an emerging market like India is just one country and, as such, represents only a very small fraction of its total revenues—hardly enough to justify senior leaders from the global center making many visits to India or investing a lot of energy to understand the Indian market’s unique features. At the same time, the local India organization does not feel empowered to develop strategy. Despite its implementer role, if the local organization proposes novel ideas for growth, the global product division is likely to rank order such ideas against proposals from all countries

and, given their unfamiliarity, the India opportunities stand little chance of funding. For a global product division, emerging markets represent a zone of discomfort, not a zone of opportunity.

Competency trap

The more an organization excels in certain competencies, the more it is likely to exploit those competencies instead of building new ones (March, 1991). Global product divisions typically have core competencies in marketing and R&D that support premium products. Emerging markets need value products—ultralow cost, functional, good-enough-quality products that demand fundamentally different competencies. Worse still, value products have the potential to make obsolete the multinational’s current competencies; they could cannibalize the sale of premium products and/or threaten the company’s brand reputation for product leadership.

Complacency trap

As the global product division succeeds with premium products, it more likely will view success as a validation of the past. This results in organizational inertia (Tripsas and Gavetti, 2000). Value products and premium products are fundamentally different and inconsistent business models; the former focus on low margin, high volume, and the latter focus on high margin, low volume. Value products involve investments today with uncertain future revenue, whereas premium products are well-tested business models with short-term payoff. Given the quarterly financial pressures, the global product division based in the DMNE’s home country would rather invest its scarce resources on premium products—its ‘known’ success formula.

The GE case study suggests a new organizational arrangement, known within the company as ‘local growth teams’ (LGTs), as a way to counter these traps (Immelt *et al.*, 2009). Building on the body of research on organizational design, future research can examine three hypotheses pertaining to LGTs. First, LGTs are likely to be full business units with their own profit and loss responsibility and dedicated local resources in all areas, including product development, sourcing, and marketing. They have to be given the power and authority to design new offerings from a blank page instead of being expected to simply adapt global products (Govindarajan and Trimble, 2005b). Second, because innovation involves experimentation and unknown outcomes,

		Where does the learning occur?	
Who is the teacher?		Emerging economy	Developed economy
	DMNEs	<u>Cell 1</u> Focus of traditional spillover literature	<u>Cell 2</u> EMNEs learning from FDI in developed countries
	EMNEs	<u>Cell 3</u> Reverse innovation: DMNEs learning from EMNEs	<u>Cell 4</u> DMNEs learning from EMNEs in developed countries

Figure 2. FDI spillover matrix

the local teams may have to focus on testing assumptions (Levinthal and March, 1993) and be judged on performance criteria other than rigid, short-term financial measures (Govindarajan and Trimble, 2010). Finally, the local teams are likely to be connected to and leverage the company's global resource base (Ambos, Asakawa, and Ambos, 2010). After all, global technology and global resources are what give DMNEs a significant edge over local competitors (Govindarajan and Trimble, 2005a; Gupta and Govindarajan, 2000).

Future research on organizational design that can help DMNEs excel simultaneously in glocalization and reverse innovation can extend and enrich our understanding of ambidextrous organizations (O'Reilly and Tushman, 2004).

Revisiting FDI spillovers

A final area where reverse innovation raises theoretical questions relates to FDI spillovers. There is a well-developed literature on how foreign firms produce spillovers for local firms in developing countries (see Figure 2), some of which are positive—e.g., helping upgrade local competitors, suppliers, and customers—and others that are negative—e.g., crowding out local firms and suppressing local technologies. (For a recent review, see Meyer and Sinani, 2009.) However, this literature could be broadened to incorporate the case of reverse innovation.

Who learns from whom and where?

One important instance of spillovers that is ignored by the mainstream literature is that of 'reverse

spillovers'—that is, learning by DMNEs about new business models, management practices, or technologies from local competitors, suppliers, and customers in emerging markets (Figure 2, cell 3). Learning in emerging economies may be a two-way street. Anecdotal evidence suggests that DMNEs usually embark on reverse innovation only when there is an actual or imminent threat of disruptive competition from local firms. An effective response to such a threat may require active and rapid learning by DMNEs from local firms about how to design, produce, sell, and service products and services for value-conscious customers. A prerequisite for such learning may be a change in the mind-set of DMNEs from that of teacher to one of humble student.

The traditional spillover literature may help us understand the underlying mechanisms for reverse spillovers. The awareness-motivation-capability framework, for instance, may explain which DMNEs learn from EMNEs and which ones do not, in the same way that it explains why some local firms learn from DMNEs and others do not (Chen, Su, and Tsai, 2007). Many DMNEs may not even be aware that they can and should be learning from local players. Further, under the glocalization approach, emerging-market subsidiaries may lack the motivation to learn locally. Finally, the rest of the organization may lack the capability to leverage such learning even if it occurs. Whereas the traditional spillover literature regards absorptive capacity in technological terms, the equivalent bottleneck for reverse spillovers may be organizational constraints that limit the absorption of ideas from emerging markets. The GE experience with ultrasound machines offers preliminary

propositions about how to create organizational absorptive capacity to capture reverse spillovers (see Appendix 1).

Reverse spillovers also can occur when EMNEs invest in developed countries and generate positive or negative spillovers for Western firms in those economies (Figure 2, cell 4). Admittedly, such investment is still limited in scope compared to investments in the other direction, but they are growing rapidly, and up to one-third of the outward foreign direct investment from emerging markets goes to developed countries, with proportions well above 50 percent for countries such as Brazil and India (Ramamurti, 2009b). It has been suggested that such investment is motivated by EMNEs' desire to obtain intangible assets, such as technology and brands, and to have a listening post in world-class clusters (Mathews, 2002; Aulakh, 2007; Luo and Tung, 2007). In other words, the traditional spillovers literature focuses almost exclusively on cell 1 of Figure 2, overlooking the possibility that EMNEs can also capture spillovers from DMNEs when they invest in developed countries (cell 2), and altogether ignores the possibility that EMNEs may be able to teach DMNEs a thing or two about serving consumers at the bottom of the pyramid (cell 3) and how to compete in mature industries (cell 4).

Thus, reverse innovation opens up several new questions about when and why reverse spillovers are likely to be important and to what extent the awareness-motivation-capability framework may also explain reverse spillovers (Meyer and Sinani, 2009). It also opens up questions about the organizational characteristics of EMNEs and DMNEs that allow some of them to capture spillovers more effectively than others.

CONCLUSIONS

Faster growth in emerging economies, combined with a 'flattening' of the world economy, is promoting innovation in poor countries. This innovation is beginning to find its way to other poor countries and sometimes to rich countries as well. This trend is led by local firms (EMNEs) and reinforced by foreign firms (DMNEs). Emerging economies seem likely to produce innovations that dramatically lower costs and add features that are especially valuable in the local context, such as portability and ease of use. From time to time, they will also produce innovations in new businesses, such as wireless communi-

cation or mobile banking, by leapfrogging past legacy technologies to frontier technologies. Some of these innovations have the potential to 'trickle up' from poor to rich countries, challenging the position of DMNEs in their traditional markets. Some DMNEs are trying to preempt this by embracing reverse innovation themselves and allowing subsidiaries in large emerging markets to pursue local-for-local or even local-for-global innovation. However, implementing such a strategy requires DMNEs to rethink their traditional organizational arrangements.

From a theoretical standpoint, reverse innovation poses many interesting questions for mainstream models of innovation, internationalization, management of DMNEs, and FDI spillovers. We have suggested a possible research agenda on reverse innovation that can be the basis for IB and global strategy scholars to revisit these models, develop arguments and theories more carefully, and test the resulting hypotheses empirically.

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Appendix 1. Implementing reverse innovation: GE's experience with ultrasound⁶

After successfully launching high-end, high-priced ultrasound machines in the United States in 1979, General Electric (GE) looked for additional markets overseas. China seemed to offer huge potential, but after more than a decade of presence there, GE's annual sales in 1995 were a mere \$5 million and growing slowly. Yet, by 2009, GE was the market leader, with a 30 percent share. How did GE make it happen?

In 1996, the company entered into a joint venture with a Chinese company, Haiying, and soon took full ownership. Subsequently, GE chose to leave the

venture with a great deal of autonomy. Over time, the unit evolved into what came to be known within GE as a 'local growth team' (LGT).

The LGT discovered that in the U.S., performance mattered most, followed by features. But in China, price mattered most, followed by portability. More than 90 percent of China's population relied on poorly funded, low-tech hospitals or basic clinics in rural villages. Because transportation was difficult, ultrasound machines had to travel to patients, not the other way around. Therefore, portability and ultralow prices were essential. As the LGT pushed ultrasound machines deeper into rural areas, it found that doctors were less familiar with ultrasounds. Its response—more training, online health guides, simpler key-boards, and built-in presets for certain tasks.

GE could not produce machines meeting these criteria while operating in the globalization mode. Its existing products could not simply be scaled down, defeatured, or adapted. It needed to create a new price-performance paradigm rather than simply shifting along the price-performance curve it already understood. GE concluded that it needed a zero-based effort to create a new price-performance curve.

Only the LGT could execute a zero-based effort. Rather than following the obvious path (i.e., trying to miniaturize existing hardware), the China LGT adopted hardware that had already been miniaturized—standard laptop computers—and then shifted most of the muscle inside an ultrasound machine from hardware to software. In 2002, GE launched its first 'compact' ultrasound machine.

The team in China remained under local control under the direction of a general manager. Its functional heads did not report to functional heads in GE Medical's global headquarters in Milwaukee. The LGT had its own profit and loss responsibility, along with the power to set its own strategy. It was a complete business managing its own value chain, including product development, manufacturing, and supply chain management. By 2008, the LGT product development team had grown fourfold, from 16 to 70 engineers; the business as a whole from 132 to 339 personnel, most recruited locally.

The team reported directly to Omar Ishrak, the leader of the global clinical systems business, a unit that included all ultrasound machines plus patient monitoring equipment. As a result, the team's resources were protected. In addition, Ishrak was able to help the LGT gain access to GE's global resource base, particularly software expertise. For

⁶ The case study is based on the first author's role as a participant-observer in GE for two years as its chief innovation consultant.

example, to the core team of 13, Ishrak added three experts from other countries, including a visionary product developer in Israel. Ishrak also oversaw frequent and productive interactions between the LGT in China and the global R&D centers, enabling the transfer of knowledge—and, in some cases, existing software modules—to the China team without constraining their efforts. He also made the leaders of the China LGT members of the Ultrasound Council, which consisted of ultrasound leaders worldwide who met three times a year to share engineering and business insights.

Ishrak evaluated the leaders of the China LGT on a broad and distinct set of criteria, not just on short-term financial outcomes and not using the same metrics used in the global business. For example, he expected faster new product introductions in China

because the government approval process for new product releases was less intricate.

As it turned out, compact ultrasounds developed for China also had a market elsewhere. This included the U.S., where they were used in unexpected ways—especially in emergency rooms and operating rooms and to guide anesthesiologists' placement of needles and catheters. Just six years after launch, compact ultrasounds were a \$278 million global product line for GE and growing at 50 to 60 percent per year.

As this case shows, reverse innovation, properly implemented, can do more than help DMNEs win in emerging markets like China. It also can reinvigorate their global product divisions and assure their continued vitality in traditional markets for years to come.